

MTH166

Lecture-2

Exact Differential Equations (EDE)

Unit 1: Ordinary Differential Equations

(Book: Advanced Engineering Mathematics by Jain and Iyengar, Chapter-4)

Topic:

Exact Differential Equations (EDE)

Learning Outcomes:

1. What is the form of Exact Differential Equations.
2. What is the Necessary and Sufficient condition for an equation to be called as Exact differential equations.
3. How to solve an Exact differential equation.

Exact Differential Equation (EDE):

An equation of the form:

$$M(x, y)dx + N(x, y)dy = 0 \quad (1)$$

is called an exact differential equation if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ (Necessary and Sufficient Condition for EDE)}$$

The solution of equation (1) is given by:

$$\int_{y=con.} M(x, y)dx + \int (Terms\ of\ N\ free\ from\ x)dy = C(\text{Constant})$$

Exact Differential Equation (EDE):

EDE Form:

$$Mdx + Ndy = 0$$

Necessary and Sufficient Condition:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution of EDE:

$$\int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x)dy = C$$

Solve the following differential equations:

Problem 1: $(x^2 - ay)dx = (ax - y^2)dy$

Solution: $(x^2 - ay)dx + (y^2 - ax)dy = 0$ (1)

Compare it with: $Mdx + Ndy = 0$

Here $M = (x^2 - ay) \Rightarrow \frac{\partial M}{\partial y} = 0 - a(1) = -a$

And $N = (y^2 - ax) \Rightarrow \frac{\partial N}{\partial x} = 0 - a(1) = -a$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so equation (1) is an exact differential equation.

Hence solution of equation (1) is given by:

$$\int_{y=\text{con.}} M dx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} (x^2 - ay) dx + \int (y^2) dy = C$$

$$\Rightarrow \frac{x^3}{3} - ay(x) + \frac{y^3}{3} = c$$

$$\Rightarrow x^3 - 3axy + y^3 = 3c$$

$$\Rightarrow x^3 - 3axy + y^3 = c_1 \text{ **Answer.** } \quad (3c = c_1)$$

Polling Quiz

The differential equation: $M(x, y)dx + N(x, y)dy = 0$ will be an exact differential equation if:

(A) $\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} = 0$

(B) $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0$ (Answer)

(C) $\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} = 0$

(D) None of these

Problem 2: $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$

Solution: $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0 \quad (1)$

Compare it with: $Mdx + Ndy = 0$

Here $M = (x^2 - 4xy - 2y^2) \Rightarrow \frac{\partial M}{\partial y} = -4x - 4y$

And $N = (y^2 - 4xy - 2x^2) \Rightarrow \frac{\partial N}{\partial x} = -4y - 4x$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so equation (1) is an exact differential equation

Hence solution of equation (1) is given by:

$$\int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} (x^2 - 4xy - 2y^2) dx + \int (y^2) dy = C$$

$$\Rightarrow \frac{x^3}{3} - 4y \left(\frac{x^2}{2} \right) - 2y^2(x) + \frac{y^3}{3} = c$$

$$\Rightarrow x^3 - 6x^2y - 6xy^2 + y^3 = 6c$$

$$\Rightarrow x^3 - 6x^2y - 6xy^2 + y^3 = c_1 \text{ **Answer.** } \quad (6c = c_1)$$

Problem 3: $(y \sin 2x)dx - (1 + y^2 + \cos^2 x)dy = 0$

Solution: $(y \sin 2x)dx - (1 + y^2 + \cos^2 x)dy = 0 \quad (1)$

Compare it with: $Mdx + Ndy = 0$

Here $M = (y \sin 2x) \Rightarrow \frac{\partial M}{\partial y} = \sin 2x$

And $N = -(1 + y^2 + \cos^2 x) \Rightarrow \frac{\partial N}{\partial x} = -(2 \cos x(-\sin x)) = \sin 2x$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so equation (1) is an exact differential equation

Hence solution of equation (1) is given by:

$$\int_{y=\text{con.}} M dx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} (y \sin 2x) dx - \int (1 + y^2) dy = C$$

$$\Rightarrow y \left(\frac{-\cos 2x}{2} \right) - y - \frac{y^3}{3} = c$$

$$\Rightarrow -3y \cos 2x - 6y - 2y^3 = 6c$$

$$\Rightarrow 3y \cos 2x + 6y + 2y^3 + c_1 = 0 \text{ **Answer.**} \quad (6c = c_1)$$

Problem 4: $(\sec x \tan x \tan y - e^x)dx + (\sec x \sec^2 y)dy = 0$

Solution: $(\sec x \tan x \tan y - e^x)dx + (\sec x \sec^2 y)dy = 0$ (1)

Compare it with: $Mdx + Ndy = 0$

Here $M = (\sec x \tan x \tan y - e^x) \Rightarrow \frac{\partial M}{\partial y} = \sec x \tan x (\sec^2 y)$

And $N = (\sec x \sec^2 y) \Rightarrow \frac{\partial N}{\partial x} = (\sec x \tan x) \sec^2 y$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so equation (1) is an exact differential equation

Hence solution of equation (1) is given by:

$$\int_{y=\text{con.}} M dx + \int (\text{Terms of } N \text{ free from } x) dy = C$$

$$\Rightarrow \int_{y=\text{con.}} (\sec x \tan x \tan y - e^x) dx + \int (0) dy = C$$

$$\Rightarrow \tan y (\sec x) - e^x = c \quad \mathbf{Answer.}$$

Polling Quiz

What is the relationship between a and b , so that the given differential equ.: $(x^2 + ay)dx + (y^2 + bx)dy = 0$ is exact:

(A) $a = 2b$

(B) $a = b$ (Answer)

(C) $a \neq b$

(D) $a \neq 2b$

